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THADOMAL SHAHANI TSEC ENGINEERING COLLEGE

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* To understand DevOps: Principles, Practices, and DevOps Engineer Role and Responsibilities

→ DevOps is a combination of cultural philosphies, practices, and tools that enables organizations to deliver applications and services at high velocity. It focuses on improving collaboration between development (Dev) and operations (Ops) teams to build, test, and release software faster and more reliably. According to the provided material, DevOps helps organizations evolved and improve products at a much faster pace compared to traditional software development methods.

Principles of DevOps -

DevOps is built on key principles that ensure efficiency and continous improvement.

1. Collaboration - Developers and operations team work together instead of in silos.
2. Communication - Open and continous communication reduces misunderstandings and delays.
3. Integration - Continous integration of code ensures early detection of errors.
4. Automation - Reduces manual effort and increases consistency in processes.
5. Continous Improvement - Encourages frequent updates and learning from failures.

The CALMS framework is widely used to represent these

principles, emphasizing teamwork, automation and data-driven decisions.

DevOps Practices-

DevOps includes several important practices that stream the software development lifecycle:

- Continous Development- Planning, coding, and version control applications.
- Continous Integration- Automated building and testing of code whenever changes are made.
- Continous Testing- Running automated tests continously to ensure software quality.
- Continous Delivery / Depevolopment - Delivering applications quickly and reliably to production.
- Infrastructure as Code (IaC) - Managing infrastructure using code for consistency.
- Configuration management - Maintaining systems in a consistent state
- Microservice Architecture - Breaking applications into smaller, manageable services.
- Continous Monitoring - Tracking performance and detecting issues in real time.

These practices help reduce errors, improve productivity, and enable faster delivery of applications.

Roles of a DevOps Engineer - A DevOps Engineer acts as a bridge between development and operations teams. Their main goal is to ensure smooth and efficient software delivery.

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Key Responsibilities -

- Designing and implementing CI/CD pipelines
- Automating build, testing, and development process.
- Managing cloud infrastructure and servers.
- Monitoring system performance and reliability.
- Ensuring security and scalability of applications.
- Handling version control and configuration management.
- Collaborating with developers, testers, and operation teams

DevOps Engineers also focus on reducing manual errors, improving system stability, and enabling faster recovery from failures.